PHYSICS REFERENCE SERIES: VOLUME I

Electromagnetism & Fundamental Field Theory — Comprehensive Module

CHAPTER 1 — SECTION 1.1

PAGE 1 — INTRODUCTION TO ELECTRIC CHARGES

1.1 Concept of Electric Charge

Electric charge is an intrinsic physical property of microscopic matter that causes it to experience a force when placed in an electromagnetic field. Along with mass, charge is one of the fundamental attributes of subatomic particles. While mass is always positive and gives rise to gravitational interaction, electric charge exists in two distinct types and gives rise to both attractive and repulsive forces.

> **Definition — Electric Charge:** Electric charge (q or Q) is a fundamental scalar property of matter that determines its electromagnetic interactions. The SI unit of electric charge is the Coulomb (C).

1.2 Historical Background and Discovery

The awareness of static electricity dates back to ancient Greece around 600 BCE, when Thales of Miletus observed that amber rubbed with fur attracted lightweight objects such as dried leaves or feathers. The modern understanding developed significantly in the 18th century:

- **William Gilbert (1600):** Coined the term "electricus" (from the Greek word *elektron* meaning amber) to describe materials that attract light objects after rubbing.
- **Charles François de Cisternay du Fay (1733):** Discovered that electrification takes two forms, which he termed "vitreous" (glass rubbed with silk) and "resinous" (amber rubbed with fur).
- **Benjamin Franklin (1747):** Introduced the modern convention of calling these two types **positive charge** and **negative charge**. Franklin established that like charges repel each other, whereas opposite charges attract each other.

1.3 Types of Charge and Atomic Structure

At the atomic scale, matter is composed of protons, neutrons, and electrons:

- **Protons:** Carry a positive charge of magnitude $e = 1.602 \times 10^{-19}\text{ C}$.
- **Electrons:** Carry a negative charge of magnitude $-e = -1.602 \times 10^{-19}\text{ C}$.
- **Neutrons:** Carry zero net charge (electrically neutral).

1.4 Role in Electromagnetism

Electric charge is the ultimate source of all electromagnetic phenomena. Stationary electric charges create electric fields, while moving electric charges (electric currents) create magnetic fields. Accelerated electric charges emit electromagnetic radiation. Thus, charge serves as the fundamental coupling parameter in Maxwell's equations.

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PAGE 2 — PROPERTIES OF ELECTRIC CHARGE

2.1 Core Physical Properties

Electric charge exhibits three basic fundamental properties that govern all physical phenomena in classical electrodynamics: Additivity, Conservation, and Quantization.

2.2 Additivity of Charges

Electric charge is a scalar quantity. If a system contains n point charges $q_1, q_2, q_3, ..., q_n$, the total charge of the system is the algebraic sum of the individual charges, taking proper account of their signs.

$$Q_{\text{total}} = \sum_{i=1}^{n} q_i = q_1 + q_2 + q_3 + ... + q_n$$

2.3 Law of Conservation of Charge

The total electric charge of an isolated physical system remains strictly constant over time. Charge can neither be created nor destroyed; it can only be transferred from one object to another or created/annihilated in equal positive-negative pairs.

> **Conservation Law:** In any closed, isolated physical process, the algebraic sum of all electric charges remains perfectly invariant.

2.4 Quantization of Charge

Electric charge is not continuous; it exists in discrete packets. Any free macroscopic or microscopic charge q in nature is always an integral multiple of the fundamental unit of charge e.

$$q = \pm n e \quad \text{where } n \in \mathbb{Z}^+ \text{ and } e = 1.602176 \times 10^{-19}\text{ C}$$

> **Conceptual Example 2.1:** Can a body carry a net charge of $4.0 \times 10^{-19}\text{ C}$?
> *Solution:* Using $q = n e \implies n = \frac{q}{e} = \frac{4.0 \times 10^{-19}}{1.602 \times 10^{-19}} = 2.497$. Since n is not an integer, such a charge cannot exist in isolation.

2.5 Summary Table of Fundamental Units

Particle	Symbol	Charge (Coulombs)	Rest Mass (kg)
Electron	e^-	$-1.602 \times 10^{-19}\text{ C}$	$9.109 \times 10^{-31}\text{ kg}$
Proton	p^+	$+1.602 \times 10^{-19}\text{ C}$	$1.672 \times 10^{-27}\text{ kg}$
Neutron	n^0	0 C	$1.674 \times 10^{-27}\text{ kg}$

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PAGE 3 — CONDUCTORS, INSULATORS, AND METHODS OF CHARGING

3.1 Classification of Materials

Based on their electrical conductivity and internal mobile charge density, materials are classified into three primary categories:

- ***Conductors:*** *Materials that allow electric charge (free electrons) to move easily through them. Examples include metals (copper, aluminum, silver), human bodies, and saline water.*
- ***Insulators (Dielectrics):*** *Materials that offer high resistance to charge movement due to tightly bound electrons. Examples include glass, rubber, wood, mica, and dry plastic.*
- ***Semiconductors:*** *Materials whose conductivity lies between conductors and insulators (e.g., Silicon, Germanium). Their conductivity increases with temperature.*

3.2 Methods of Electrification

Objects can acquire a net electric charge through three primary physical mechanism methods:

> ***1. Charging by Friction (Triboelectric Effect):*** *When two dissimilar insulating materials are rubbed together, thermal friction supplies sufficient energy to transfer electrons from the material with lower work function to the one with higher work function.*

> ***2. Charging by Conduction:*** *Direct contact between a charged body and an uncharged conductor allows electric charge to flow between them until an equilibrium potential state is established.*

> ***3. Charging by Induction:*** *A process where a neutral conductor acquires a charge without direct physical contact with a charged object. Bringing a charged rod near an uncharged insulated conductor redistributes internal free charges; grounding the far side leaves a net opposite charge.*

3.3 Comparison of Charging Methods

Method	Contact Required?	Resulting Sign of Charge	Applicable Materials
Friction	*Yes (rubbing)*	*Equal and opposite on both*	*Insulators*
Conduction	*Yes (touching)*	*Same sign as charging body*	*Conductors*
Induction	*No (proximity)*	*Opposite sign to charging body*	*Conductors*

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PAGE 4 — COULOMB'S LAW

4.1 Statement of Coulomb's Law

Formulated by Charles-Augustin de Coulomb in 1785, Coulomb's Law quantifies the electrostatic force exerted between two stationary, non-moving point charges in a vacuum or free space.

> ***Coulomb's Law Statement:*** *The magnitude of the electrostatic force between two point electric charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance separating their centers.*

4.2 Mathematical Formulation

For two point charges q_1 and q_2 separated by a distance r in a vacuum, the magnitude of force F is:

$$F = k_e \frac{|q_1 q_2|}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{|q_1 q_2|}{r^2}$$

4.3 Physical Constants and Units

- k_e *is the Electrostatic Constant (or Coulomb Constant):* $k_e \approx 8.98755 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2 \approx 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.
- ε_0 *is the Permittivity of Free Space (vacuum permittivity):* $\varepsilon_0 \approx 8.854187 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2) \text{ or F/m}$.

4.4 Influence of the Medium

When charges are embedded in a material medium rather than a vacuum, the force is reduced by a factor equal to the dielectric constant (K or ε_r) of that medium:

$$F_{\text{medium}} = \frac{F_{\text{vacuum}}}{K} = \frac{1}{4\pi\varepsilon} \frac{|q_1 q_2|}{r^2} \quad \text{where } \varepsilon = \varepsilon_0 \varepsilon_r = \varepsilon_0 K$$

> ***Example 4.1:*** *Calculate the electrostatic force between two point charges of $+2\ \mu\text{C}$ and $+6\ \mu\text{C}$ placed 3 cm apart in a vacuum.*
> *Solution: Here $q_1 = 2 \times 10^{-6}\text{ C}$, $q_2 = 6 \times 10^{-6}\text{ C}$, and $r = 0.03\text{ m}$.*
> $F = \frac{(9 \times 10^9) \times (2 \times 10^{-6}) \times (6 \times 10^{-6})}{(0.03)^2} = \frac{0.108}{0.0009} = 120\text{ N (repulsive)}$.

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PAGE 5 — COULOMB'S LAW IN VECTOR FORM

5.1 Vector Representation of Force

Since force is a vector quantity possessing both magnitude and direction, expressing Coulomb's Law in vector notation is crucial for analyzing complex 2D and 3D charge configurations.

> ***Vector Equation:*** *Let $\vec{r}_1$ and $\vec{r}_2$ be the position vectors of point charges q_1 and q_2 relative to an origin O. The force $\vec{F}_{21}$ exerted on charge q_2 by charge q_1 is given by:*

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\vec{r}_2 - \vec{r}_1|^2} \hat{r}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\vec{r}_{21}|^3} \vec{r}_{21}$$

where $\vec{r}_{21} = \vec{r}_2 - \vec{r}_1$ is the displacement vector directed from q_1 to q_2, and $\hat{r}_{12} = \frac{\vec{r}_{21}}{|\vec{r}_{21}|}$ is the unit vector along that line.

5.2 Sign Conventions and Direction

- *If $q_1 q_2 > 0$ (like charges), $\vec{F}_{21}$ points along $\hat{r}_{12}$ (repulsive force pushing q_2 away from q_1).*
- *If $q_1 q_2 < 0$ (opposite charges), $\vec{F}_{21}$ points opposite to $\hat{r}_{12}$ (attractive force pulling q_2 toward q_1).*

5.3 Newton's Third Law Compatibility

The electrostatic force exerted by q_2 on q_1 is $\vec{F}_{12}$. Since $\vec{r}_{12} = -\vec{r}_{21}$, it follows directly that:

$$\vec{F}_{12} = -\vec{F}_{21}$$

This confirms that Coulomb electrostatic forces obey Newton's Third Law of Motion (action and reaction are equal in magnitude and opposite in direction) and act strictly along the central line joining the charge centers.

CHAPTER 1 — SECTION 1.6

PAGE 6 — SUPERPOSITION PRINCIPLE

6.1 System of Multiple Charges

Coulomb's Law specifically defines the force between two isolated point charges. When an array of multiple charges is present, experimental evidence proves that electrostatic forces obey the Principle of Superposition.

> ***Principle of Superposition:*** *The net electrostatic force acting on any target charge due to a system of surrounding charges is equal to the vector sum of individual forces exerted on it by each charge evaluated independently. The force between any pair of charges is unaffected by the presence of other charges.*

6.2 Mathematical Formulation for N Charges

Consider a system of N discrete point charges $q_1, q_2, \dots, q_N$ *located at position vectors* $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N$. *The net force* $\vec{F}_1$ *acting on charge* q_1 *is:*

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1N} = \frac{q_1}{4\pi\varepsilon_0} \sum_{i=2}^{N} \frac{q_i}{|\vec{r}_1 - \vec{r}_i|^3} (\vec{r}_1 - \vec{r}_i)$$

> ***Example 6.1 (Equilateral Triangle System):*** *Three equal positive charges* q *are positioned at the vertices of an equilateral triangle of side length* a. *Find the magnitude of the net force on any one charge.*
> *Solution: The target charge experiences two repulsive forces of equal magnitude* $F_0 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}$ *directed at an angle of* 60° *relative to each other.*
> $F_{\text{net}} = \sqrt{F_0^2 + F_0^2 + 2 F_0 F_0 \cos(60^\circ)} = \sqrt{3 F_0^2} = \sqrt{3} F_0 = \frac{\sqrt{3} q^2}{4\pi\varepsilon_0 a^2}$.

CHAPTER 1 — SECTION 1.7

PAGE 7 — CONTINUOUS CHARGE DISTRIBUTION

7.1 Macroscopic Charge Models

When dealing with macroscopic physical objects containing vast numbers of microscopic charges (e.g., $\sim 10^{23}$ electrons), calculating forces by summing over individual discrete charges becomes impractical. Instead, charge is modeled as a continuous distribution over lines, surfaces, or volumes.

7.2 Types of Charge Density

1. Linear Charge Density (λ): *Used when charge is uniformly distributed along a one-dimensional line or wire.*

$$\lambda = \frac{dq}{dl} \quad \text{Unit: C/m}$$

2. Surface Charge Density (σ): *Used when charge is spread over a two-dimensional surface or thin sheet.*

$$\sigma = \frac{dq}{dA} \quad \text{Unit: C/m}^2$$

3. Volume Charge Density (ρ): *Used when charge is distributed throughout a three-dimensional solid volume.*

$$\rho = \frac{dq}{dV} \quad \text{Unit: C/m}^3$$

7.3 Force Due to Continuous Charge Distributions

To compute the total force exerted by a continuous charge distribution on a test charge q_0, we integrate the infinitesimal force $d\vec{F}$ exerted by an elemental charge dq over the entire region:

$$\vec{F} = \frac{q_0}{4\pi\varepsilon_0} \int \frac{dq}{r^2} \hat{r} = \frac{q_0}{4\pi\varepsilon_0} \left(\int_L \frac{\lambda\, dl}{r^2}\hat{r} + \int_S \frac{\sigma\, dA}{r^2}\hat{r} + \int_V \frac{\rho\, dV}{r^2}\hat{r} \right)$$

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PAGE 8 — ELECTRIC FIELD CONCEPT AND PHYSICAL SIGNIFICANCE

8.1 Definition of Electric Field

The electric field is a vector field that permeates the space around any electric charge distribution. It represents the physical mediator through which charges exert forces on one another across distance without direct contact.

> ***Electric Field Intensity ($\vec{E}$):*** *Defined as the electrostatic force experienced per unit positive test charge placed at a given point in space, provided the test charge is vanishingly small ($q_0 \to 0$) so as not to distort the source distribution.*

$$\vec{E}(\vec{r}) = \lim_{q_0 \to 0} \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2} \hat{r} \quad \text{Unit: N/C or V/m}$$

8.2 Physical Significance of the Field

The concept of an electric field replaces the instantaneous "action-at-a-distance" model with a localized field-based interaction. A source charge Q modifies the properties of surrounding space by establishing a field $\vec{E}$. Any secondary test charge q_0 introduced into this field feels a local force $\vec{F} = q_0 \vec{E}$ directly at its immediate coordinate.

8.3 Electric Field of Point Charges

- *For a positive source charge ($Q > 0$), $\vec{E}$ points radially outward away from the charge.*
- *For a negative source charge ($Q < 0$), $\vec{E}$ points radially inward toward the charge.*

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PAGE 9 — ELECTRIC FIELD LINES AND PROPERTIES

9.1 Visualizing Electric Fields

Electric field lines (lines of force), introduced by Michael Faraday, are continuous pictorial curves drawn such that the tangent to the line at any point gives the direction of the electric field vector $\vec{E}$ at that location.

9.2 Fundamental Rules Governing Field Lines

1. ***Origin and Termination:*** *Field lines start on positive charges and end on negative charges. In the case of a single charge, lines extend to or terminate at infinity.*
2. ***Continuity:*** *Field lines are continuous curves in charge-free space, containing no abrupt breaks or discontinuities.*
3. ***Non-Intersection:*** *Two electric field lines can never cross each other. If they intersected, two different tangents could be drawn at the point of intersection, implying two conflicting directions for the net electric field at a single point, which is physically impossible.*
4. ***Density and Strength:*** *The relative spatial density of field lines (number of lines per unit cross-sectional area) is directly proportional to the magnitude of the electric field in that region.*
5. ***Perpendicular to Conductors:*** *Electric field lines always meet the surface of a conductor at right angles (90°) under electrostatic conditions.*
6. ***No Closed Loops:*** *Electrostatic field lines do not form closed loops because electrostatic fields are conservative in nature.*

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PAGE 10 — ELECTRIC DIPOLE AND DIPOLE MOMENT

10.1 Definition of an Electric Dipole

*An electric dipole consists of a pair of equal and opposite point charges (+**q** and **-q**) separated by a small distance **2a**.*

> ***Electric Dipole Moment (\vec{p}):*** *A vector quantity measuring the strength and orientation of an electric dipole. Its magnitude is equal to the product of the charge magnitude **q** and the separation vector **2\vec{a}**.*

$$\vec{p} = q \times (2\vec{a}) \quad \text{Unit: Coulomb-meter (C}\cdot\text{m)}$$

Direction Convention: *By physical convention, the vector \vec{p} points strictly from the **negative charge (-q) toward the positive charge (+q)** along the dipole axis.*

10.2 Molecular Dipoles

Molecules are categorized into two structural classes based on charge symmetry:

- ***Non-polar Molecules:*** *The centers of positive and negative charge coincide (e.g., \text{H}_2, \text{O}_2, \text{CO}_2). They possess zero permanent electric dipole moment in the absence of an external field.*
- ***Polar Molecules:*** *Centers of positive and negative charge are permanently separated due to asymmetric molecular geometry (e.g., \text{H}_2\text{O}, \text{HCl}). They possess a permanent non-zero electric dipole moment.*

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PAGE 11 — ELECTRIC FIELD DUE TO A DIPOLE (AXIAL AND EQUATORIAL)

11.1 Field on the Axial Line (End-On Position)

*Consider a point **P** situated on the extended axis of a dipole at distance **r** from its center **O**.*

$$\vec{E}_{\text{axial}} = \frac{1}{4\pi\varepsilon_0} \frac{2 \vec{p} r}{(r^2 - a^2)^2}$$

For a short dipole ($r \gg a$):

$$\vec{E}_{\text{axial}} \approx \frac{1}{4\pi\varepsilon_0} \frac{2\vec{p}}{r^3} \quad \text{(Parallel to } \vec{p}\text{)}$$

11.2 Field on the Equatorial Line (Broadside-On Position)

*Consider a point **Q** located on the equatorial perpendicular bisector of the dipole at distance **r** from center **O**.*

$$\vec{E}_{\text{equatorial}} = -\frac{1}{4\pi\varepsilon_0} \frac{\vec{p}}{(r^2 + a^2)^{3/2}}$$

For a short dipole ($r \gg a$):

$$\vec{E}_{\text{equatorial}} \approx -\frac{1}{4\pi\varepsilon_0} \frac{\vec{p}}{r^3} \quad \text{(Antiparallel to } \vec{p}\text{)}$$

11.3 Comparison Ratio

*For a short dipole at equal distances **r** from the dipole center:*

$$|\vec{E}_{\text{axial}}| = 2\,|\vec{E}_{\text{equatorial}}|$$

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PAGE 12 — TORQUE AND POTENTIAL ENERGY OF A DIPOLE IN A UNIFORM FIELD

12.1 Torque on a Dipole

When an electric dipole $\vec{p}$ is placed in a uniform external electric field $\vec{E}$ making an angle θ with the field direction, equal and opposite forces $\vec{F} = \pm q\vec{E}$ act on its ends, forming a couple.

$$\vec{\tau} = \vec{p} \times \vec{E} \implies \tau = p E \sin\theta$$

- **Stable Equilibrium ($\theta = 0^\circ$):** Torque $\tau = 0$. $\vec{p}$ is parallel to $\vec{E}$.
- **Unstable Equilibrium ($\theta = 180^\circ$):** Torque $\tau = 0$. $\vec{p}$ is antiparallel to $\vec{E}$.
- **Maximum Torque ($\theta = 90^\circ$):** $\tau_{\text{max}} = p E$.

12.2 Potential Energy of the Dipole

The work done W in rotating the dipole against the field torque from angle θ_1 to θ_2 is stored as electrostatic potential energy U:

$$U(\theta) = -\vec{p} \cdot \vec{E} = -p E \cos\theta$$

CHAPTER 1 — SECTION 1.13

PAGE 13 — DIPOLE IN A NON-UNIFORM ELECTRIC FIELD

13.1 Net Force in Non-Uniform Fields

In a non-uniform electric field, the magnitude of \vec{E} varies spatially across the length of the dipole. Consequently, the forces acting on +q and -q are unequal (\vec{F}_+ \neq -\vec{F}_-), resulting in both a net translational force and a rotational torque.

$$\vec{F}_{\text{net}} = (\vec{p} \cdot \nabla) \vec{E}$$

13.2 Direction of Motion

- *When \vec{p} is aligned parallel to \vec{E}, the net force pulls the dipole toward regions of **stronger electric field**.*
- *When \vec{p} is aligned antiparallel to \vec{E}, the net force pushes the dipole toward regions of **weaker electric field**.*

> ***Physical Phenomenon Example:*** *An electrified plastic comb attracts uncharged bits of paper. The non-uniform field of the comb induces dipoles in paper molecules, pulling them toward the comb where the field gradient is strongest.*

CHAPTER 1 — SECTION 1.14

PAGE 14 — ELECTRIC FLUX

14.1 Concept of Electric Flux

Electric flux measures the total number of electric field lines passing orthogonally through a given surface area element.

> ***Definition — Electric Flux (\Phi_E):*** *The surface integral of the electric field vector \vec{E} over a specified surface area S.*

$$\Phi_E = \iint_S \vec{E} \cdot d\vec{A} = \iint_S E \cos\theta \, dA \quad \text{Unit: N}\cdot\text{m}^2/\text{C or V}\cdot\text{m}$$

where $d\vec{A} = \hat{n}\, dA$ is the vector area element directed outward along the surface normal unit vector $\hat{n}$, and θ is the angle between $\vec{E}$ and $\hat{n}$.

14.2 Sign Conventions for Flux

- ***Inward Flux ($\theta > 90^\circ$):*** *Negative flux ($\Phi_E < 0$) enters a closed volume.*
- ***Outward Flux ($\theta < 90^\circ$):*** *Positive flux ($\Phi_E > 0$) leaves a closed volume.*

CHAPTER 1 — SECTION 1.15

PAGE 15 — GAUSS'S LAW

15.1 Fundamental Statement of Gauss's Law

Gauss's Law is one of Maxwell's fundamental equations governing electrodynamics. It provides a direct relation between electric flux through a closed surface and the net charge enclosed within that surface.

> ***Gauss's Law Statement:*** *The total net electric flux* Φ_E *passing through any arbitrary closed surface (termed a Gaussian Surface) is equal to* $1/\varepsilon_0$ *times the net enclosed electric charge* Q_{enclosed}.

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

15.2 Key Properties and Features

1. ***Surface Independence:*** *The flux depends solely on total enclosed charge, not on the shape, geometry, or size of the Gaussian surface.*
2. ***External Charges:*** *Charges outside the Gaussian surface contribute zero net flux over the entire closed surface because lines entering the surface must also exit it.*

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PAGE 16 — APPLICATION OF GAUSS'S LAW: INFINITELY LONG STRAIGHT WIRE

16.1 Field Calculation for an Infinite Line Charge

Consider an infinitely long, straight wire carrying a uniform linear charge density λ.

16.2 Gaussian Surface Selection

By cylindrical symmetry, the electric field $\vec{E}$ is everywhere radial and perpendicular to the wire. Construct a coaxial cylindrical Gaussian surface of radius r and length L.

16.3 Mathematical Derivation

The cylindrical surface consists of three sections: two flat circular end caps and one curved cylindrical surface.

- *Flat End Caps: $\vec{E} \perp d\vec{A} \implies \vec{E} \cdot d\vec{A} = 0$. Flux is zero.*
- *Curved Surface: $\vec{E} \parallel d\vec{A} \implies \int \vec{E} \cdot d\vec{A} = E (2\pi r L)$.*

Applying Gauss's Law:

$$E (2\pi r L) = \frac{\lambda L}{\varepsilon_0} \implies E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

CHAPTER 1 — SECTION 1.17

PAGE 17 — APPLICATION OF GAUSS'S LAW: UNIFORM INFINITE PLANE SHEET

17.1 Field Calculation for a Plane Sheet

Consider an infinite, non-conducting, thin plane sheet carrying a uniform surface charge density σ.

17.2 Gaussian Surface Selection

By planar symmetry, $\vec{E}$ is directed normally away from the sheet on both sides. Construct a cylindrical Gaussian "pillbox" of cross-sectional area A piercing through the sheet.

17.3 Derivation

- *Curved Surface: $\vec{E} \perp d\vec{A} \implies \text{Flux} = 0$.*
- *Two Flat Face Caps: $\vec{E} \parallel d\vec{A} \implies \text{Flux} = E A + E A = 2 E A$.*

Applying Gauss's Law:

$$2 E A = \frac{\sigma A}{\varepsilon_0} \implies E = \frac{\sigma}{2\varepsilon_0}$$

Crucial Insight: *The electric field produced by an infinite plane sheet of charge is completely uniform and entirely independent of distance r from the sheet.*

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PAGE 18 — APPLICATION OF GAUSS'S LAW: THIN SPHERICAL SHELL

18.1 Field due to a Uniform Thin Spherical Shell

Consider a thin spherical shell of radius R carrying a total charge Q distributed uniformly over its surface with surface charge density $\sigma = \frac{Q}{4\pi R^2}$.

18.2 Case 1: Point Outside the Shell ($r > R$)

Construct a concentric spherical Gaussian surface of radius $r > R$.

$$E\,(4\pi r^2) = \frac{Q}{\varepsilon_0} \implies E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

Conclusion: For external points, a uniformly charged spherical shell behaves exactly as if its entire charge were concentrated as a point charge at its center.

18.3 Case 2: Point Inside the Shell ($r < R$)

Construct a concentric spherical Gaussian surface of radius $r < R$. Since all charge resides on the exterior shell surface, $Q_{\text{enclosed}} = 0$.

$$E\,(4\pi r^2) = 0 \implies E = 0$$

Conclusion: The electric field inside a thin uniformly charged spherical shell is identically zero everywhere.

CHAPTER 1 — SECTION 1.19

PAGE 19 — CONDUCTORS IN ELECTROSTATIC EQUILIBRIUM

19.1 Fundamental Electrostatic Properties of Conductors

When electric charges on a conductor are stationary, the conductor is in electrostatic equilibrium, giving rise to six fundamental properties:

> ***1. Zero Internal Field:*** *The electric field inside a conductor is zero everywhere ($\vec{E}_{\text{inside}} = 0$). Any internal mobile electrons redistribute until all internal fields are neutralized.*

> ***2. Surface Field Orthogonality:*** *Just outside a conductor, $\vec{E}$ is strictly perpendicular to the surface at every point ($E_n = \frac{\sigma}{\varepsilon_0}$).*

> ***3. Zero Internal Net Charge:*** *Any excess charge resides entirely on the exterior surface of a conductor.*

> ***4. Constant Potential:*** *The electrostatic potential is constant throughout the entire volume of a conductor and equal to its value on the surface.*

19.2 Electrostatic Shielding

The field inside any cavity inside a conductor is zero, regardless of external field strengths. This effect, known as Electrostatic Shielding, protects sensitive instruments inside hollow metal enclosures (Faraday Cages).

CHAPTER 1 — SUMMARY & ASSESSMENT

PAGE 20 — FORMULA SUMMARY AND CONCEPTUAL REVIEW

20.1 Master Equation Reference Table

Physical Concept	Mathematical Formula	Key Parameters
Quantization of Charge	$q = n\,e$	$e = 1.602 \times 10^{-19}\text{ C}$
Coulomb's Law	$F = \frac{1}{4\pi\varepsilon_0}\frac{\|q_1\,q_2\|}{r^2}$	$\varepsilon_0 = 8.854 \times 10^{-12}\text{ F/m}$
Electric Field (Point Charge)	$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$	Radially outward (+) or inward (-)
Dipole Moment	$\vec{p} = q\,(2\vec{a})$	Directed from $-q$ to $+q$
Axial Dipole Field ($r \gg a$)	$E_{\text{axial}} = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$	Parallel to dipole moment $\vec{p}$
Equatorial Dipole Field	$E_{\text{eq}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}$	Antiparallel to dipole moment $\vec{p}$
Torque on Dipole	$\vec{\tau} = \vec{p} \times \vec{E}$	Maximum at $\theta = 90^\circ$
Gauss's Law	$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$	Valid for any closed surface

20.2 Key Retrieval Concepts for RAG Q&A

- ***Conservation:*** *Total net charge in an isolated system is strictly constant.*
- ***Quantization:*** *Free charge is restricted to discrete integer multiples of elementary charge e.*
- ***Inverse Square Behavior:*** *Field and force decay with $1/r^2$ for point charges, but $1/r^3$ for electric dipoles.*
- ***Infinite Sheet Field:*** *Uniform field independent of distance ($E = \frac{\sigma}{2\varepsilon_0}$).*